Hybrid Physics–ML Based EAQKD System with Explainable Attack Detection and Adaptive Parameter Optimization

Summer Internship and Research Experience-II submitted in partial fulfilment for the requirement of the degree of Bachelor of Technology in Computer Science and Engineering

by

Bibekananda Routray

Regd.No.:2302040001

Computer Science and Engineering

Under the Supervision of

Dr. Alina Dash,

Dr. Satya Prakash Sahoo



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VEER SURENDRA SAI UNIVERSITY OF TECHNOLOGY,
BURLA, SAMBALPUR – 768018

[2025-2026]



Department of Computer Science and Engineering

Veer Surendra Sai University of Technology, Burla, Odisha

Date: November 6, 2026

Certificate of Examination

Name: Bibekananda Routray

Registration Number: 2302040001

Title of the Project: Hybrid Physics–ML Based EAQKD System with Explainable Attack Detection and Adaptive Parameter Optimization

This is to certify that the report entitled "**Hybrid Physics–ML Based EAQKD System with Explainable Attack Detection and Adaptive Parameter Optimization**" submitted by **Mr. Bibekananda Routray** Regd.no. **2302040001** of the Department of Computer Science and Engineering, Veer Surendra Sai University of Technology, Odisha has been examined by me. I am satisfied with the volume, quality and correctness of the work.

Dr. Satya Prakash Sahoo
Assistant Professor,
Department Of CSE,
VSSUT, BURLA

Declaration

I hereby declare that the project titled "**Hybrid Physics–ML Based EAQKD System with Explainable Attack Detection and Adaptive Parameter Optimization"** submitted to Veer Surendra Sai University of Technology for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a result of original work carried out in this report. I also declare that the work has not been submitted, in whole or in part, to any other university as an exercise for a degree or any other qualification.

Date: November 6, 2026
VSSUT, BURLA

Bibekananda Routray
Regd.No.:2302040001

Acknowledgment

I would like to thank our supervisor, **Dr. Alina Dash** and **Dr. Satya Prakash Sahoo** for their invaluable guidance, constant support, patience, and constructive suggestions, all of which greatly contributed to the success of our technical seminar.

I would like to thank the HOD, Department of Computer Science and Engineering for his support to carry out our project work. We would like to thank all the faculty and staff members of the Department of Computer Science and Engineering at Veer Surendra Sai University of Technology, Burla, for their help and support.

I am deeply thankful to the Vice-Chancellor for his moral support and for providing us with all the necessary facilities during our candidature.

I would like to thank our friends and family members for their unconditional love and support.

Name: Bibekananda Routray
Registration No.: 2302040001

Abstract

This project presents a Hybrid Physics–Machine Learning based EAQKD simulation system designed for secure and intelligent quantum communication. The system simulates the process of Entanglement-Based Authenticated Quantum Key Distribution (EAQKD), where secret quantum keys are generated and shared securely between two communicating users using the principles of quantum mechanics. The project studies the effect of different communication conditions such as channel noise, photon loss, transmission distance, and disturbances on the security and performance of the system. Important performance parameters such as Quantum Bit Error Rate (QBER) and secure key generation rate are analyzed during the simulation. To improve security analysis, the proposed system integrates machine learning techniques for intelligent attack detection. The machine learning model analyzes multiple parameters including QBER, noise level, photon loss, and distance to identify possible eavesdropping attacks more accurately than traditional fixed-threshold methods. Explainable AI (XAI) is also included so that the system can provide understandable reasons behind attack prediction results, improving transparency and user trust. The project further includes leakage risk detection and security score evaluation for analyzing the overall security condition of the communication channel. The generated quantum key is also used with AES-based encryption to demonstrate practical secure message communication. In addition, a web-based interactive dashboard is developed to visualize system behavior, performance metrics, attack status, leakage risk, and security score in a simple and user-friendly manner. The proposed framework provides an adaptive, intelligent, and easy-to-understand platform for studying modern secure quantum communication systems under different operating conditions.

Keywords: Quantum Key Distribution, EAQKD, Machine Learning, Explainable AI, QBER, Hybrid Model, Security Optimization

Table of Contents

1 Introduction 1
1.1. Problem Statement 1
1.2. Objectives 2
1.3. Motivation 2
2. Literature Review 3
3. Background Study 6
3.1. Quantum Key Distribution 6
3.2. Entanglement-Based Quantum Key Distribution Protocols 7
3.3. Security Challenges in Practical QKD Systems 8
3.4. AES-Based Secure Communication Using Quantum Keys 8
3.5. Simulation-Based Analysis of EAQKD Systems 9
3.6. Machine Learning in QKD Systems 9
3.7. Intelligent Security Analysis and Adaptive Optimization in QKD 9
4. Conclusion and Future Scopes 11
4.1. Conclusion 11
4.2. Future Scope 11
5. References 12

List of Figures

Figure 1: Illustration of Quantum Key Distribution 6

Figure 2: Illustration of Entanglement-Based Quantum Key Distribution 7

Figure 3: Quantum Key Distribution System 10

List of Abbreviations

QKD: Quantum Key Distribution
EAQKD: Entanglement-Based Authenticated Quantum Key Distribution
QBER: Quantum Bit Error Rate
ML: Machine Learning
XAI: Explainable Artificial Intelligence
AES: Advanced Encryption Standard
RSA: Rivest–Shamir–Adleman
ECC: Elliptic Curve Cryptography
IBM: International Business Machines
AI: Artificial Intelligence
E91: Ekert 1991 Quantum Key Distribution Protocol
BB84: Bennett and Brassard 1984 Quantum Key Distribution Protocol

1 Introduction

With the rapid growth of digital communication, cloud computing, and online data exchange, the need for secure communication systems has increased significantly. Most modern communication systems use classical cryptographic methods such as RSA and AES for protecting sensitive information. However, the development of quantum computing has created serious security concerns because quantum algorithms such as Shor's algorithm can break many traditional public-key encryption systems [1]. Quantum Key Distribution (QKD) has emerged as a promising solution because it uses the principles of quantum mechanics for secure key generation and communication [2]. Among different QKD approaches, Entanglement-Based Authenticated Quantum Key Distribution (EAQKD) provides strong security using quantum entanglement and authenticated communication channels.

Despite its advantages, practical EAQKD systems still face several security challenges such as channel noise, photon loss, detector imperfections, transmission errors, and possible side-channel attacks, which increase the Quantum Bit Error Rate (QBER) and reduce communication reliability [3]. Traditional QKD systems mainly use fixed threshold methods for attack detection, which are not reliable under changing communication conditions. To address these challenges, this project proposes a Hybrid Physics–Machine Learning based EAQKD simulation system that combines quantum communication simulation, machine learning-based attack detection, Explainable AI, leakage risk analysis, security score evaluation adaptive optimization, and AES-based secure communication. The proposed system also includes a web-based dashboard for visualization of communication performance and security parameters.

1.1. Problem Statement

Traditional communication systems mainly depend on mathematical encryption methods such as RSA and ECC for secure communication. However, with the advancement of quantum computing, these methods may become vulnerable to future quantum attacks [1]. Although Quantum Key Distribution (QKD) provides theoretically secure communication using quantum mechanics, practical QKD systems still face many challenges such as channel noise, photon loss, detector imperfections, transmission disturbances, and eavesdropping attacks [3]. These problems increase the Quantum Bit Error Rate (QBER) and reduce secure key generation performance. Most existing QKD systems also use fixed threshold methods for attack detection and parameter selection, which are not suitable for dynamic communication environments. In addition, traditional systems do not provide proper leakage risk analysis, adaptive optimization, or explainable decision-making. Therefore, there is a need for an intelligent and adaptive EAQKD simulation system that can analyze communication conditions, detect attacks accurately, evaluate leakage risk and security score, optimize system parameters dynamically, and provide understandable prediction results for secure and reliable quantum communication.

1.2. Objectives

This project presents a Hybrid Physics–Machine Learning based EAQKD simulation system for secure and intelligent quantum communication. The proposed system simulates entanglement-based quantum key distribution under different communication conditions such as noise, photon loss, and transmission distance, and integrates AES-based secure message encryption using the generated quantum key. The system uses machine learning techniques for intelligent eavesdropping attack detection by analyzing parameters such as QBER, noise level, photon loss, and distance. Explainable AI is included to provide understandable prediction results and improve system transparency. The project also performs leakage risk detection, security score evaluation, and adaptive parameter optimization for improving communication security, reliability, and secure key generation performance. In addition, a web-based dashboard is developed to visualize system performance and security information in an interactive manner.

1.3. Motivation

With the rapid growth of quantum computing and digital communication, the security of traditional encryption methods is becoming a major concern. Classical cryptographic systems such as RSA may become vulnerable to future quantum attacks. Although Quantum Key Distribution (QKD) provides secure communication using quantum mechanics, practical EAQKD systems still face problems such as noise, photon loss, and difficulty in detecting attacks correctly. Most existing systems use fixed threshold methods, which do not work well in changing communication conditions. They also lack proper leakage analysis and clear explanation of prediction results. Machine learning and Explainable AI can help in improving intelligent attack detection and adaptive decision-making. These challenges motivated the development of a Hybrid Physics–Machine Learning based EAQKD simulation system for secure, adaptive, and intelligent quantum communication.

2. Literature Review

Entanglement-Based Quantum Key Distribution (EAQKD) systems have become an important research area because they provide secure communication using the principles of quantum mechanics. Researchers have developed different QKD protocols, simulation models, adaptive optimization techniques, and intelligent security methods to improve secure key generation and communication reliability under practical conditions such as channel noise, photon loss, and transmission errors. Recent studies have also explored machine learning, leakage risk analysis, and Explainable AI techniques for improving intelligent attack detection and adaptive security analysis in modern quantum communication systems.

Portmann et al. 2021 [1] studied the theoretical and practical security aspects of quantum cryptography systems. Their work explained the importance of authentication, side-channel protection, and secure implementation in practical QKD communication. The research also showed that improper implementation and leakage in communication channels can reduce the overall security of QKD systems.

Xu et al. 2021 [2] introduced an adaptive optimization framework for QKD systems operating under changing communication conditions. Their method dynamically adjusted communication parameters to improve secure key generation performance and communication stability. The study achieved lower transmission errors and better adaptability compared to traditional fixed-parameter systems.

Bacco et al. 2022 [3] proposed a software-defined simulation framework for quantum communication networks. Their framework analyzed communication efficiency, secure key generation, and network scalability under different operating conditions. The study demonstrated that simulation-based analysis is useful for developing practical and large-scale QKD communication systems.

Kaur et al. 2022 [4] studied the effects of channel noise and photon loss in practical QKD systems. Their research analyzed how environmental disturbances increase the Quantum Bit Error Rate (QBER) and reduce secure communication performance. The study emphasized the importance of proper noise management and channel optimization for maintaining reliable communication.

Ali et al. 2023 [5] explored Explainable AI techniques for intelligent decision-making systems. Their research focused on improving transparency, interpretability, and user trust in machine learning models. The study explained how Explainable AI can help users understand prediction results more clearly in intelligent security applications.

Sharma et al. 2023 [6] proposed a machine learning-based attack detection model for QKD systems using communication parameters such as QBER, photon loss, and noise level. Their work improved attack detection accuracy compared to traditional threshold-based methods. The research showed that machine learning can provide better adaptive security analysis under dynamic communication conditions.

Mariani et al. 2024 [7] developed a simulation of the E91 entanglement-based QKD protocol using IBM Quantum and Qiskit tools. Their work modeled entanglement generation, polarization measurement, error correction, and privacy amplification under different noise conditions. The study analyzed how communication disturbances affect QBER and secure key generation performance.

Zhang et al. 2024 [8] proposed an adaptive optimization method for improving secure communication performance in quantum systems. Their method dynamically adjusted communication parameters according to changing channel conditions. The research improved communication reliability, secure key generation efficiency, and overall system performance.

Dabiri et al. 2025 [9] studied the physical limitations of entanglement-based satellite QKD systems for long-distance communication. Their research analyzed the effects of beam divergence, background noise, and tracking errors on secure key generation performance. The study showed that long-distance communication conditions significantly affect communication reliability and QBER.

Al-Kuwari et al. 2025 [10] explored quantum machine learning-based attack detection methods for QKD systems. Their work analyzed communication parameters such as QBER, entropy, and photon loss for intelligent eavesdropping detection. The study demonstrated that intelligent models can improve security analysis and attack detection performance compared to traditional methods.

Mohamed et al. 2026 [11] introduced the EAQKD protocol, which combined entanglement-based quantum communication with authenticated classical communication. Their work used simulation-based analysis to evaluate QBER, secure key generation rate, and communication performance under realistic communication conditions. The study showed that the proposed protocol can maintain secure communication with acceptable performance and reliability.

Sarkar et al. 2026 [12] studied side-channel leakage in entanglement-based QKD systems. Their research showed that even small amounts of leaked information can reduce the security of practical quantum communication protocols. The study also discussed the scalability limitations and security challenges of large quantum communication networks.

Weinberg et al. 2026 [13] proposed an Explainable AI framework called Q-SHAP for quantum systems. The framework used SHAP-based interpretability techniques to explain prediction and decision-making processes in quantum intelligent systems. Their work improved transparency, interpretability, and understanding of machine learning predictions in quantum communication applications.

Table 1 : Literature Survey

Paper	Year	Techniques Used	Remarks
Portmann et al. [1]	2021	Authentication methods, side-channel protection, theoretical QKD security	Improved secure implementation and practical security analysis
Xu et al. [2]	2021	Adaptive optimization framework, dynamic parameter adjustment	Improved communication stability and reduced transmission errors
Bacco et al. [3]	2022	Software-defined QKD simulation, scalable communication networks	Improved simulation-based analysis for large-scale QKD systems
Kaur et al. [4]	2022	Noise analysis, photon loss modeling, QBER evaluation	Studied the effects of environmental disturbances on communication performance
Ali et al. [5]	2023	Explainable AI (XAI), intelligent decision-making systems	Improved transparency and interpretability in machine learning systems
Sharma et al. [6]	2023	Machine learning-based attack detection, QBER and noise analysis	Improved attack detection accuracy under dynamic conditions
Mariani et al. [7]	2024	IBM Quantum simulation, Qiskit, E91 protocol, privacy amplification	Simulated entanglement generation and analyzed QBER under noise
Zhang et al. [8]	2024	Adaptive optimization, communication parameter tuning	Improved secure key generation efficiency and communication reliability
Dabiri et al. [9]	2025	Satellite-based EAQKD, photon-level modeling, free-space optical communication	Long-distance communication affected by beam divergence and tracking errors
Al-Kuwari et al. [10]	2025	Quantum machine learning, intelligent eavesdropping detection	Improved attack detection using QBER, entropy, and photon loss analysis
Mohamed et al. [11]	2026	EAQKD protocol, authenticated classical communication, discrete-event simulation	Maintained secure communication with acceptable QBER values
Sarkar et al. [12]	2026	Side-channel leakage analysis, entanglement-based QKD security	Small leaked information reduced security and network scalability
Weinberg et al. [13]	2026	Q-SHAP, Explainable AI, SHAP-based interpretability techniques	Improved transparency and understanding of quantum decision-making

3. Background Study

3.1. Quantum Key Distribution

Quantum Key Distribution (QKD) is a secure communication method that uses the principles of quantum mechanics to generate and share a secret key between two users. Unlike classical cryptography, QKD provides security based on the laws of physics instead of mathematical complexity. The BB84 protocol and E91 protocol are among the most widely used QKD protocols for secure quantum communication.

In QKD systems, the secret key is generated by transmitting photons through a quantum channel. The sender, usually called Alice, sends photons with different quantum states or polarizations, while the receiver, called Bob, measures them using selected measurement bases. After transmission, Alice and Bob compare part of their measurement information, such as the used bases, through a classical communication channel without revealing the actual key values. The matching measurement results are then used to generate a shared secret key. If an eavesdropper tries to intercept the photons, the quantum states become disturbed, which increases the Quantum Bit Error Rate (QBER) and helps detect possible attacks. A low QBER indicates secure communication, while a high QBER may indicate channel disturbance or eavesdropping attempts.

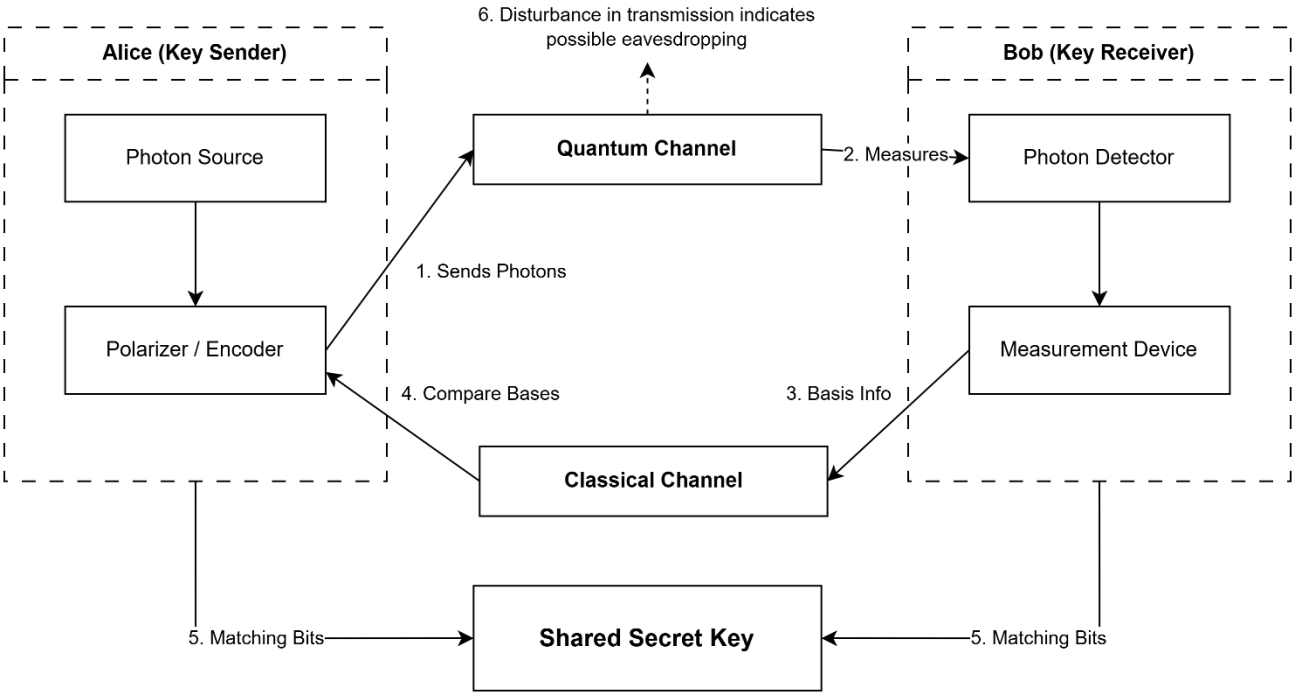


Figure 1 : Illustration of Quantum Key Distribution

3.2. Entanglement-Based Quantum Key Distribution Protocols

Entanglement-Based Quantum Key Distribution (EAQKD) uses pairs of entangled photons to establish secure communication between users. In these systems, entangled photons are generated from an entangled photon source and transmitted to the sender and receiver through a quantum channel. The sender and receiver measure the received photons using selected measurement bases. After transmission, both users compare the used bases through a classical communication channel without revealing the actual measurement values. The matching measurement results are then used to generate the shared secret key.

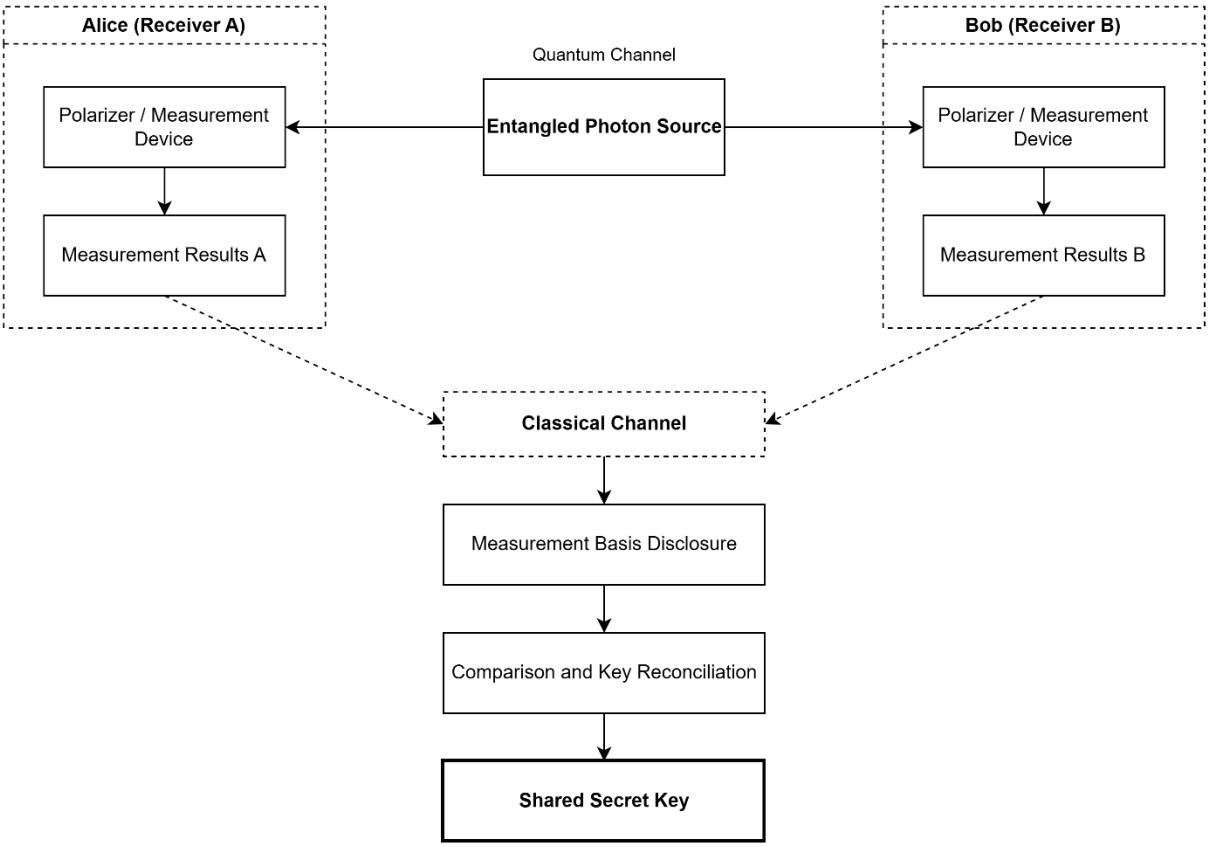


Figure 2: Illustration of Entanglement-Based Quantum Key Distribution

Entanglement-based protocols provide strong theoretical security because any attempt to measure or intercept the quantum states disturbs the communication channel.
A simple entangled quantum state commonly used in this protocol is represented as:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \qquad (1)$$

In this equation, $|\Phi^+\rangle$represents one of the four Bell states used in quantum communication. The term $\frac{1}{\sqrt{2}}$ is a normalization factor that ensures the total probability of the quantum state remains equal to one. The states $|00\rangle$ and $|11\rangle$ represent the correlated states of two entangled photons. In the $|00\rangle$state, both photons have the value 0, while in the $|11\rangle$ state, both photons have the value 1. The plus sign (+) indicates that the system exists in a superposition of both states simultaneously. Because of this entanglement property, measuring one photon immediately determines the state of the other photon, even when they are separated by large distances.

Recent research has used IBM Quantum and Qiskit tools for simulating entanglement generation, polarization measurement, error correction, and privacy amplification under different communication conditions. However, practical implementation of EAQKD systems still faces challenges such as photon loss, detector inefficiencies, and channel noise, which reduce communication efficiency and increase the Quantum Bit Error Rate (QBER).

3.3. Security Challenges in Practical QKD Systems

Practical QKD systems face several challenges in real communication environments. Factors such as photon loss, environmental noise, detector imperfections, transmission errors, and channel attenuation can increase the Quantum Bit Error Rate and reduce secure key generation performance. Long-distance communication systems are also affected by beam divergence, background noise, and tracking errors. In addition, side-channel leakage and improper authentication methods can reduce the overall security of practical QKD systems. These challenges make it difficult to distinguish between normal channel disturbances and actual eavesdropping attacks. Therefore, reliable security analysis and adaptive communication techniques are important for maintaining secure quantum communication.

3.4. AES-Based Secure Communication Using Quantum Keys

In practical QKD systems, the generated quantum secret key is commonly used with classical encryption algorithms such as AES for secure message communication. AES is a symmetric encryption algorithm that provides fast and secure data encryption. In hybrid communication systems, QKD is mainly used for secure key generation and distribution, while AES is used for encrypting the actual message data.
AES encryption can be represented as:

$$C = E_K(P) \qquad (2)$$

where $P$is the plaintext message, $K$is the secret quantum-generated key, and $C$is the encrypted ciphertext.
This combination improves both communication security and practical implementation efficiency. In the proposed project, AES encryption is integrated with the quantum-generated secret key to

demonstrate secure communication between users.

3.5. Simulation-Based Analysis of EAQKD Systems

Simulation plays an important role in QKD research because practical quantum communication hardware is expensive and difficult to implement. Simulation frameworks allow researchers to study protocol behaviour under different communication conditions without using physical quantum devices. Recent studies have used simulation tools to analyze entanglement generation, photon transmission, error correction, privacy amplification, and secure key generation under different noise conditions. Simulation-based analysis also helps researchers evaluate attack scenarios, optimize communication parameters, and improve system reliability. Because of these advantages, simulation has become an important approach for developing and testing modern EAQKD systems.

3.6. Machine Learning in QKD Systems

Machine Learning (ML) is widely used for intelligent analysis and decision-making in modern communication systems. In Quantum Key Distribution systems, ML techniques help analyze communication parameters such as QBER, noise level, photon loss, and transmission distance for detecting abnormal behavior and possible eavesdropping attacks. Unlike traditional fixed-threshold methods, machine learning models can learn from data and adapt to changing communication conditions. Recent research has shown that ML-based methods can improve attack detection accuracy, communication reliability, and adaptive optimization in practical QKD systems. Machine learning is also useful for leakage risk analysis and security score evaluation in intelligent quantum communication systems.

3.7. Intelligent Security Analysis and Adaptive Optimization in QKD

Traditional QKD systems mainly use fixed threshold values such as predefined QBER limits for attack detection. However, fixed-rule systems are not reliable in changing communication environments where channel conditions continuously vary. Recent research has explored machine learning techniques for analyzing parameters such as QBER, noise level, photon loss, and transmission distance for intelligent attack detection. Explainable AI techniques are also used to provide understandable reasons behind prediction results and improve system transparency. Modern QKD research additionally focuses on leakage risk detection, security score evaluation, and adaptive parameter optimization for improving communication reliability and secure key generation performance under dynamic channel conditions.

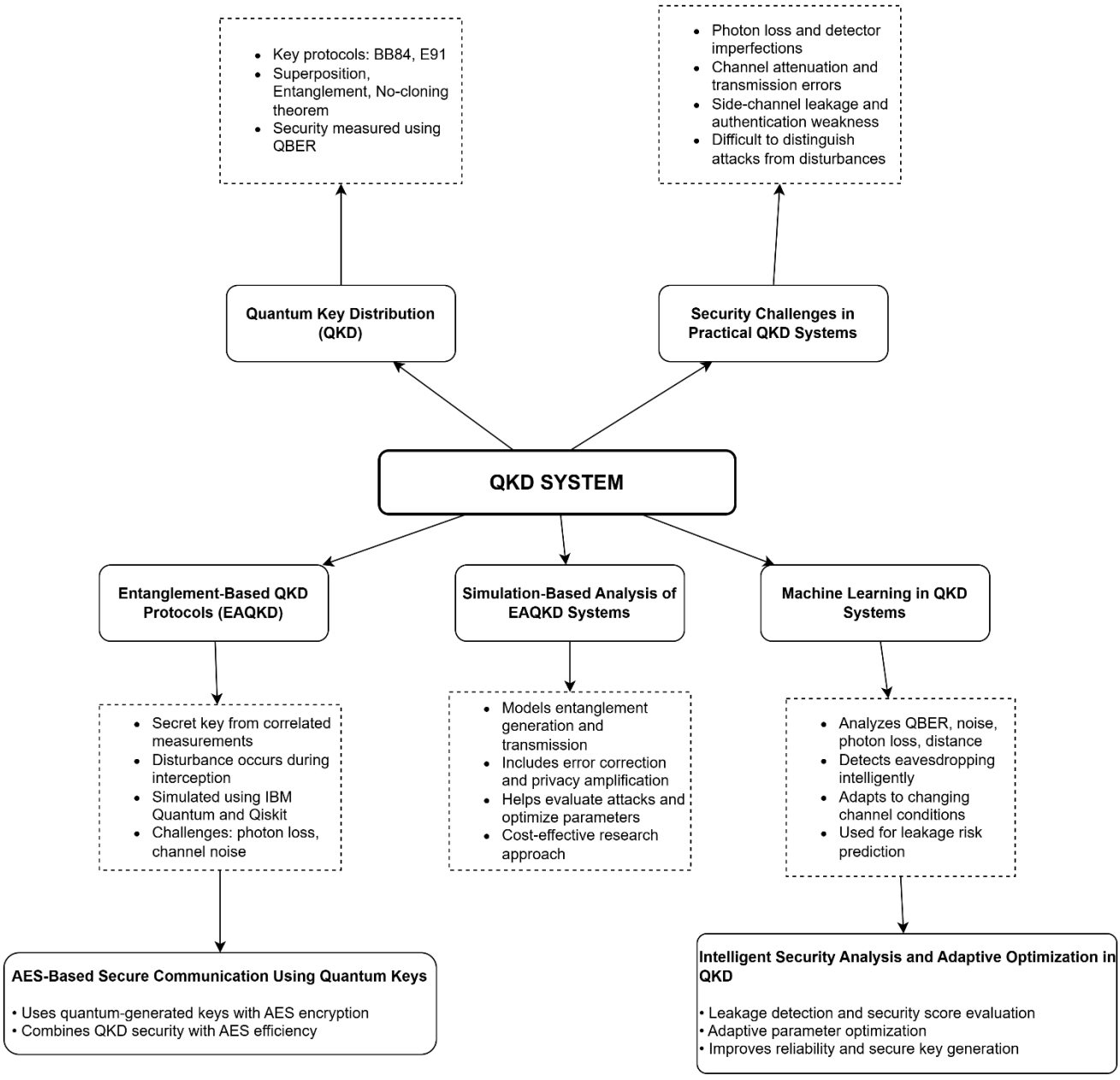


Figure 3 : Quantum Key Distribution System

4. Conclusion and Future Scopes

4.1. Conclusion

This project presented a Hybrid Physics–Machine Learning based EAQKD simulation system for secure quantum communication. The system simulated entanglement-based quantum key distribution under different conditions such as noise, photon loss, and distance. Machine learning was used for intelligent attack detection, while Explainable AI helped in understanding prediction results. The project also included leakage risk detection, security score evaluation, adaptive optimization, and AES-based secure communication. A web-based dashboard was developed for visualizing system performance and security parameters. Overall, the project provides an intelligent and secure platform for studying modern quantum communication systems.

4.2. Future Scope

Future work can focus on extending the proposed system to real-world quantum communication environments by using real experimental data and quantum hardware platforms. This can help improve the accuracy and reliability of the simulation results. Advanced machine learning techniques such as deep learning and reinforcement learning can also be added for better attack detection and automatic parameter optimization.

Further research may include more advanced leakage risk detection methods and stronger security mechanisms such as post-quantum cryptographic authentication. The system can also be expanded for satellite-based communication and large-scale quantum networks using quantum repeaters. In addition, improving computational efficiency, real-time monitoring, and visualization features can make the system more practical and suitable for future secure quantum communication technologies.

5. References

[1] C. Portmann and R. Renner. *Security in quantum cryptography*. arXiv preprint arXiv:2102.00021, 2021.

[2] Y. Xu, Z. Li, and H. Guo. *Adaptive optimization techniques for practical quantum key distribution systems*. IEEE Access, 9:112345–112356, 2021.

[3] M. Bacco, A. Gotta, and M. Marchese. *Software-defined simulation framework for quantum communication networks*. Future Internet, 14(2):45–58, 2022.

[4] H. Kaur and A. Kaur. *Analysis of noise and photon loss in practical quantum key distribution systems*. International Journal of Quantum Information, 20(4):2250012, 2022.

[5] S. Ali, M. Ahmed, and R. Khan. *Explainable artificial intelligence for intelligent decision-making systems*. Journal of Artificial Intelligence Research, 76:455–478, 2023.

[6] R. Sharma, P. Verma, and S. Gupta. *Machine learning-based attack detection in quantum key distribution systems*. IEEE Transactions on Information Forensics and Security, 18:2550–2562, 2023.

[7] L. Mariani, L. Salatino, C. Attanasio, S. Pagano, and R. Citro. *Simulation of an entanglement-based quantum key distribution protocol*. European Physical Journal Plus, 139:602, 2024.

[8] X. Zhang, Y. Chen, and T. Wang. *Adaptive optimization for secure quantum communication systems*. Quantum Information Processing, 23(5):188, 2024.

[9] M. T. Dabiri, M. Hasna, S. Al-Kuwari, and K. Qaraqe. *Physical limits of entanglement-based quantum key distribution over long-distance satellite links*. arXiv preprint arXiv:2506.20798, 2025.

[10] S. Al-Kuwari, N. Mohamed, and A. Rashid. *Quantum machine learning-based attack detection for QKD systems*. International Journal of Quantum Communication, 7(1):15–29, 2025.

[11] N. Mohamed and S. Al-Kuwari. *EAQKD: Entanglement-Based Authenticated Quantum Key Distribution*. arXiv preprint arXiv:2603.02375, 2026.

[12] S. Sarkar, T. Prasad, and K. Horodecki. *Entanglement is not sufficient for most practical entanglement-based QKD protocols*. arXiv preprint arXiv:2603.06400, 2026.

[13] A. I. Weinberg. *Q-SHAP: Explainable AI for Quantum Circuits via Quantum-Native and Hybrid Classical Approaches*. ResearchGate preprint, 2026.